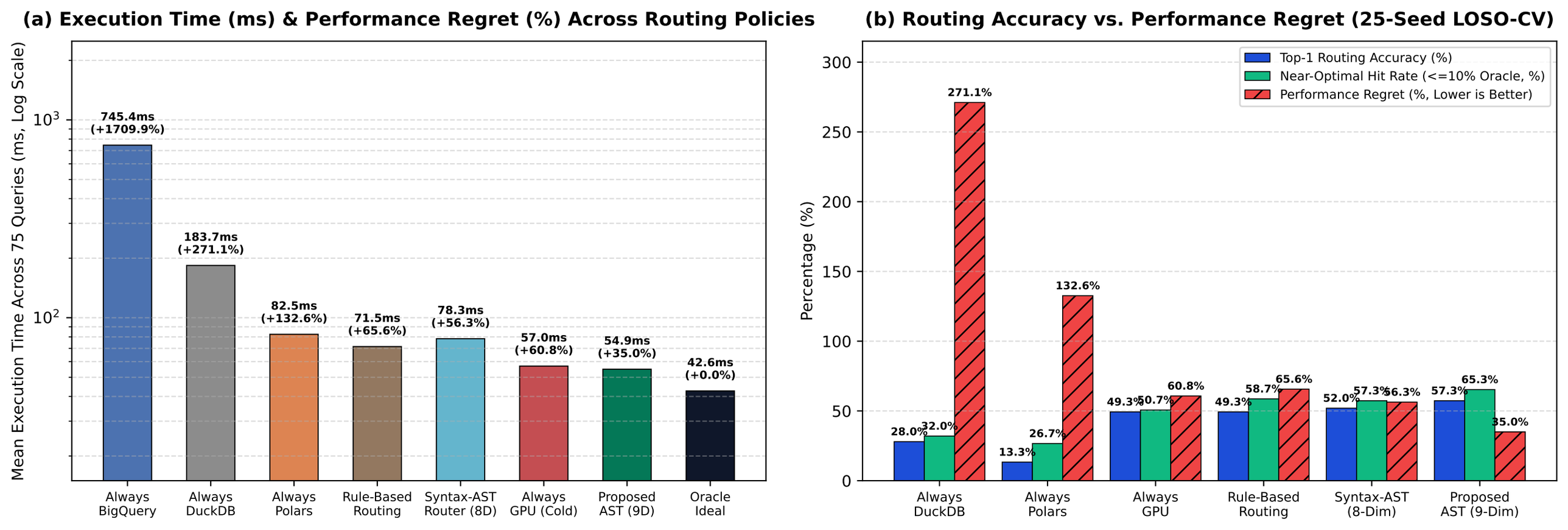
(a) Execution Time (ms) & Performance Regret (%) Across Routing Policies
Mean Execution Time Across 75 Queries (ms, Log Scale)
10^3
10^2
745.4ms (+1709.9%)
183.7ms (+271.1%)
82.5ms (+132.6%)
71.5ms (+65.6%)
78.3ms (+56.3%)
57.0ms (+60.8%)
54.9ms (+35.0%)
42.6ms (+0.0%)
Always BigQuery
Always DuckDB
Always Polars
Rule-Based Routing
Syntax-AST Router (8D)
Always GPU (Cold)
Proposed AST (9D)
Oracle Ideal
(b) Routing Accuracy vs. Performance Regret (25-Seed LOSO-CV)
Top-1 Routing Accuracy (%)
Near-Optimal Hit Rate (<=10% Oracle, %)
Performance Regret (%, Lower is Better)
Percentage (%)
0
50
100
150
200
250
300
28.0%
32.0%
271.1%
13.3%
26.7%
132.6%
49.3%
50.7%
60.8%
49.3%
58.7%
65.6%
52.0%
57.3%
56.3%
57.3%
65.3%
35.0%
Always DuckDB
Always Polars
Always GPU
Rule-Based Routing
Syntax-AST (8-Dim)
Proposed AST (9-Dim)